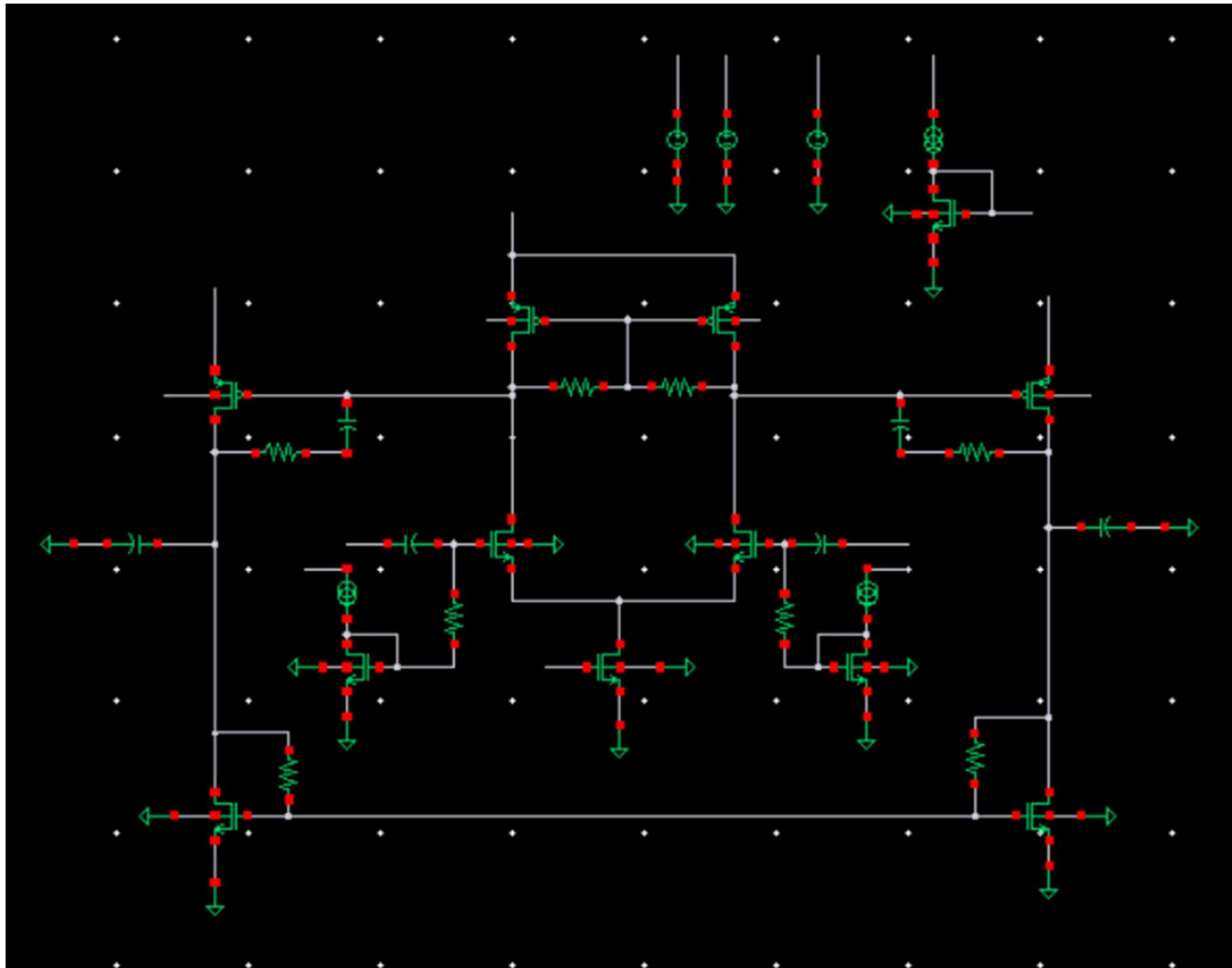
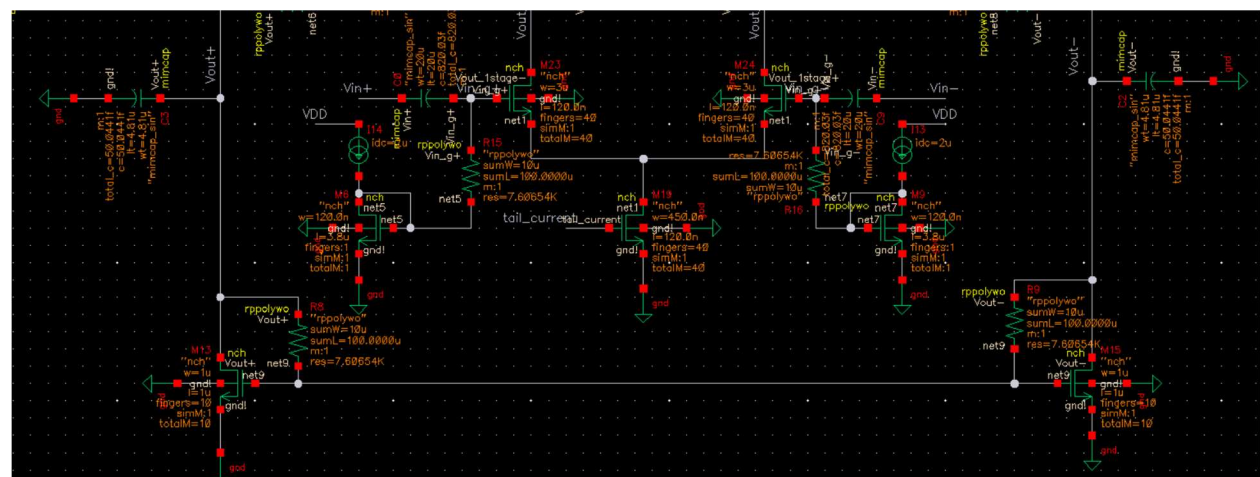


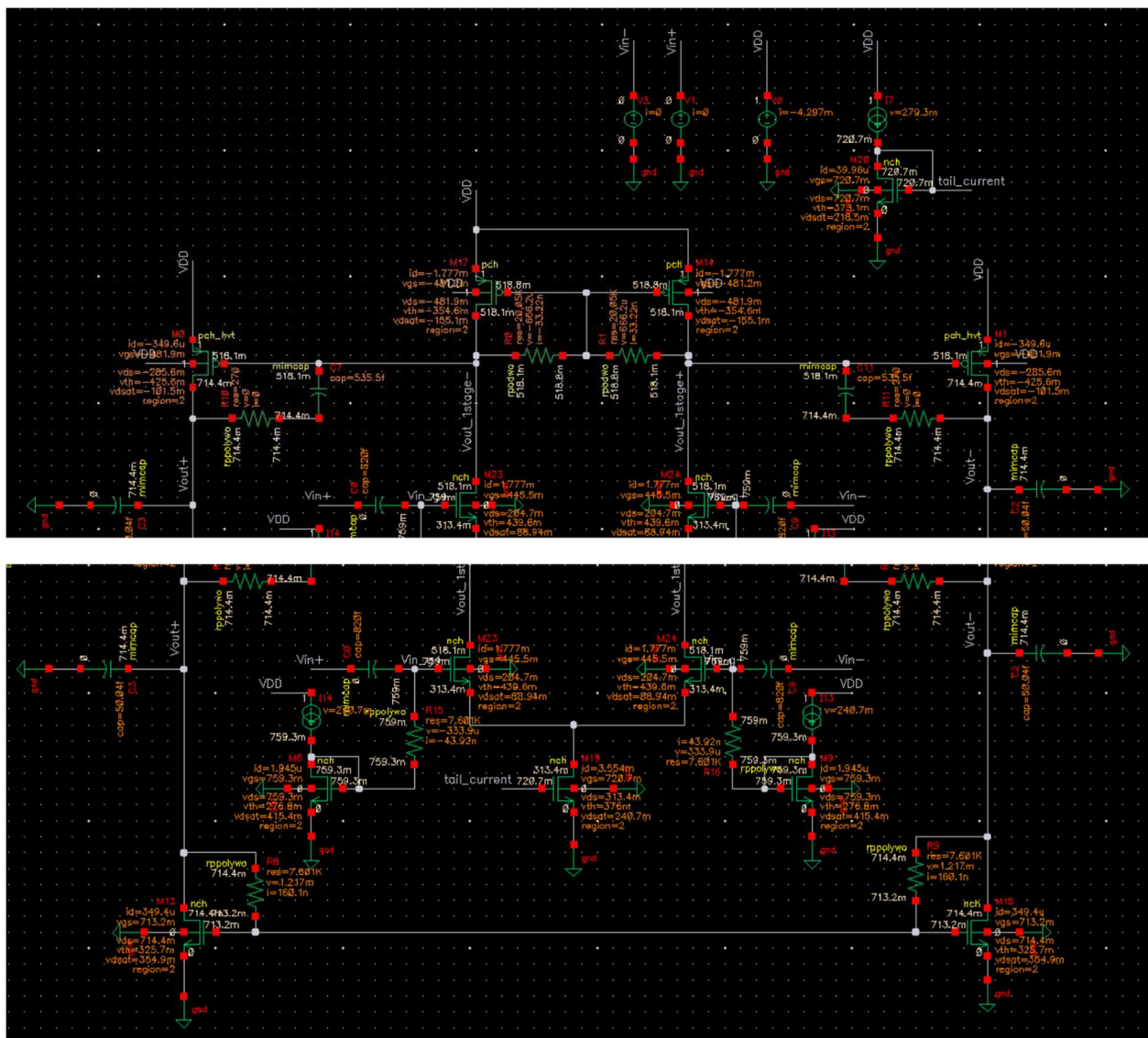
	Designed Circuit's Spec	Target
Gain	41.3 dB	> 40dB
Unity Gain Frequency	5.66 GHz	> 5 GHz
Phase Margin	51.1 degrees	> 60 degrees
Power Consumption	4.3 mW	< 4.5 mW
Pk-Pk Amplitude (Single-Ended)	544 mV	> 400mV

As shown in the figure below, common-mode feedbacks were implemented twice in the designed circuit.

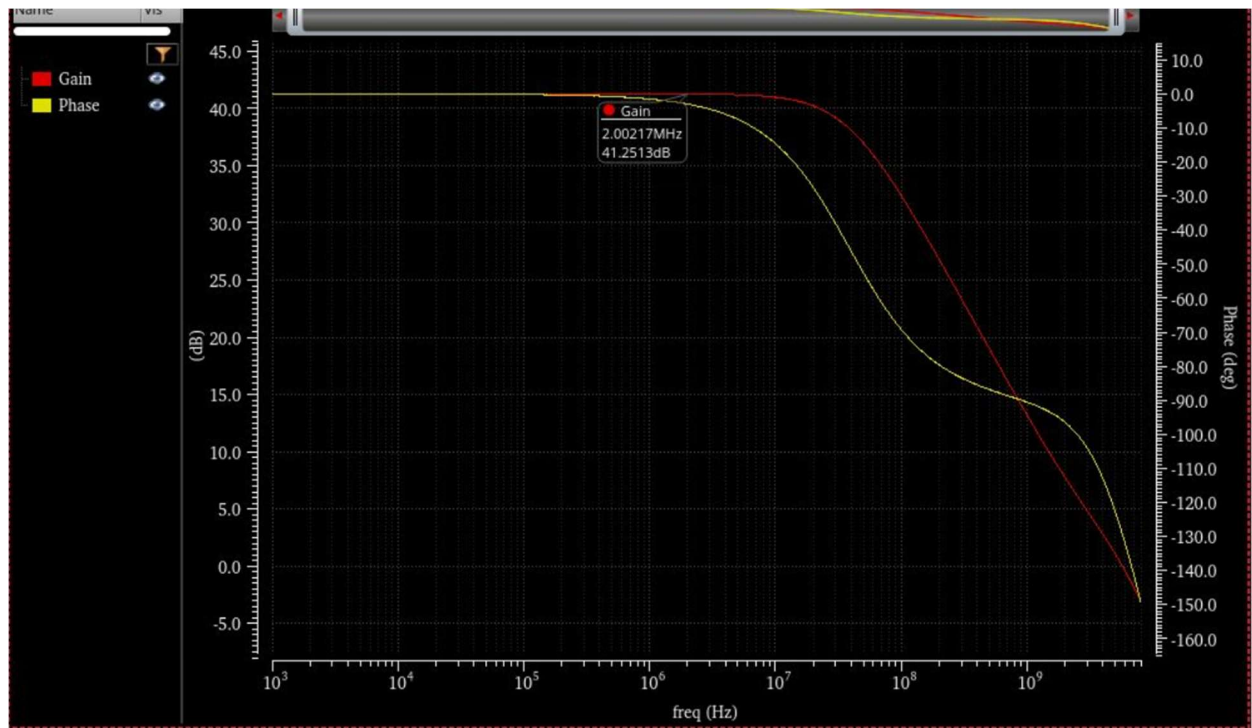
Overall Design



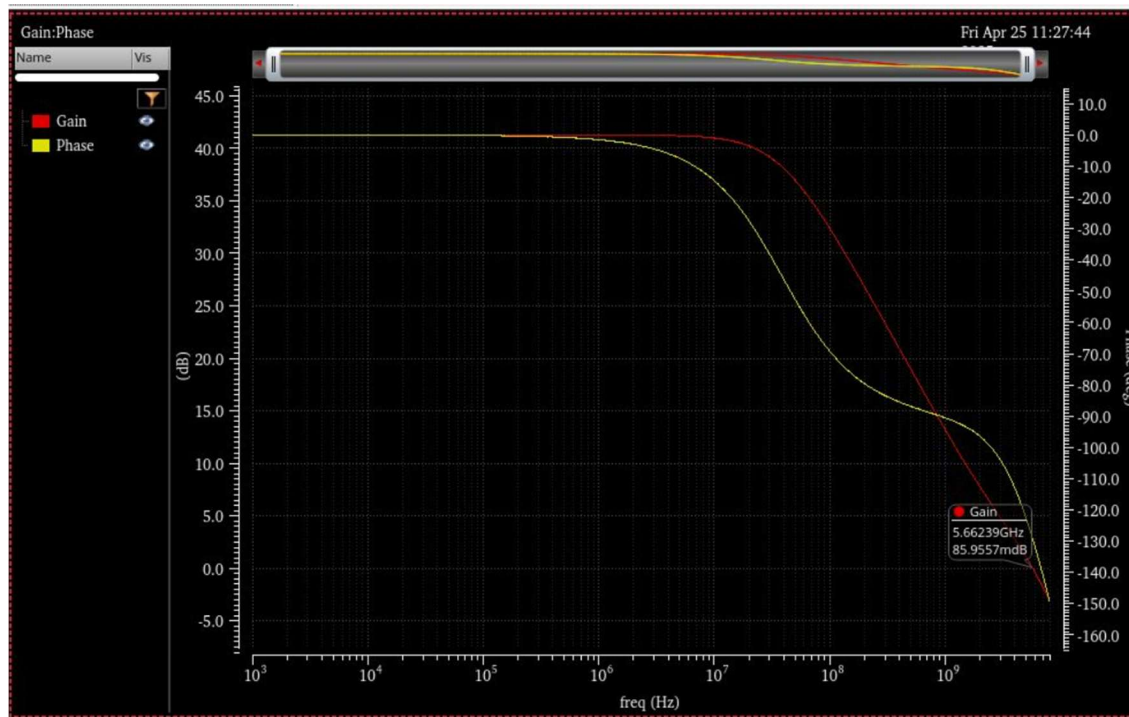




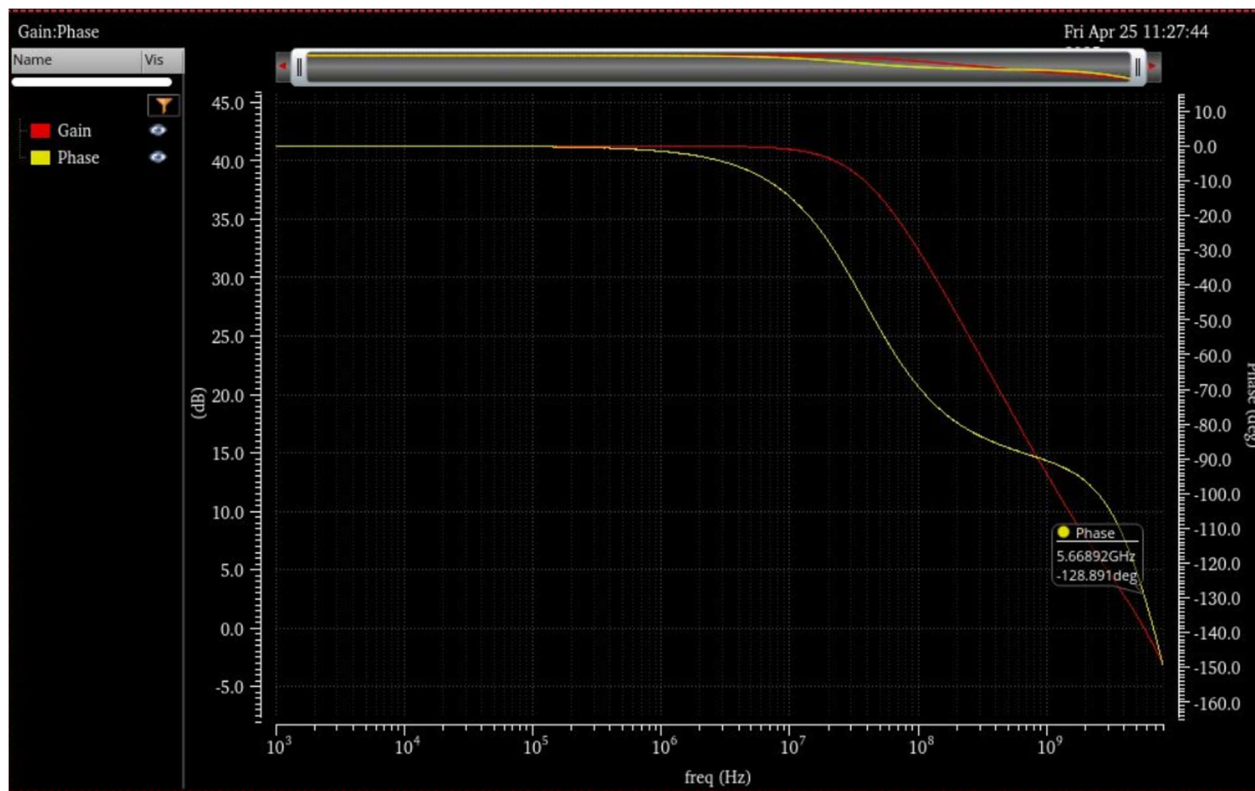
Gain at low = 41.3 dB



Unity Gain Frequency = 5.66 GHz



Phase Margin = $180 - 128.89 = 51.11$ degrees



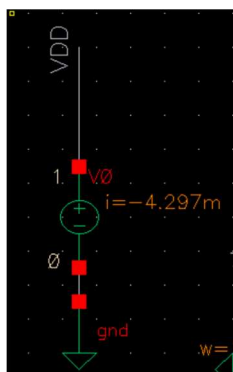
Peak-to-Peak Voltage at the Output Node (Single Ended)

$$= V_{dd} - V_{d,sat} \text{ of } M_{13} - V_{d,sat} \text{ of } M_0$$

$$= 1 \text{ V} - 354.9 \text{ mV} - 101.5 \text{ mV}$$

$$= 543.6 \text{ mV}$$

$$\text{Power Consumption} = 4.297 \text{ W}$$



DRC Error

Cell/Rule	Color	Count	Level
TOP VLSI_Midterm_layout		15	0
✗ DRM.R.1		1	
✗ M1.DN.1		1	
✗ M2.DN.1		1	
✗ M3.DN.1		1	
✗ M4.DN.1		1	
✗ M5.DN.1		1	
✗ M6.DN.1		1	
✗ M7.DN.1		1	
✗ M8.DN.1		1	
✗ M8.DN.5:L		1	
✗ M9.DN.2		1	
✗ OD.DN.2		1	
✗ PO.DN.2		3	

LVS Error

